

Brazilian E-Commerce Public Dataset

Data Curation by Ahmad Assi

General Dataset Information:

File Name:	archive
Description:	Orders at Olist Store from 2016 to 2018
Dataset Details:	100k Rows & (52 Columns at 9 files csv)
Size:	42.6 MB (44,717,580 bytes)
Source:	Kaggle - Dataset Link

Data Profile:

- Imported the dataset into an **DuckDb** database using **DBeaver**.
- Examine dataset through **SQL** queries to understand its structure.
- Query essential data relevant to the problem statements.
- Exported the dataset to Excel for further analysis.
- Applied **Excel filters** to identify patterns and inconsistencies.
- Analyzed the unique values of each attribute for irregularities.

Notable Features (Prior to Export)

- **99,441** Unique Order, the order may have more than one seller or product.
- **112,650** Unique items sold.
- **3,095** Unique seller, Large orders for each one.
- **72** Unique product categories, Large orders for each category .
- **27** Unique states and **4,119** Unique cities, with large order per region.
- **There are 8 order statuses:**
 - approved, Canceled, created, delivered, invoiced, processing, shipped, unavailable
- **5 payment types:**

- boleto, credit_card, debit_card, voucher, not_defined
- compare **review score** with The difference in days from (**order_purchase_timestamp** and **order_delivered_customer_date**) .
- **Price**: A possible gauge for potential revenues.

Recorded Inconsistency:

- many orders contain **unknown product and seller name and price** (null value).
- Several orders with **no reviews score**.
- many orders contain **unknown product category name** .
- Some values are written in **scientific notation** .
- Some orders have an **unknown order delivery date**.
- One order contains an **unknown payment type (Null value)**.
- One seller has **numerical fields within text strings (seller state)**.
- Exaggerated Customer wait delivery time are suspected outliers and may be test listings.

● **Data Wrangling:**

- Using **SQL**, extracted **21 columns** from **9 tables** by applying different JOIN types to merge them into 5 main tables: **Orders, Order Items, Customers, Products, Sellers**.
- The payment_type table was excluded as it is **not relevant** to the analysis.

Column Name	Number of Null Values	Action Made	Reason
product_name	595	Fill it with "unnamed_product"	Maintain data consistency
review_score	646	Fill it with the Mean	Preserve data integrity for analysis

order_delivered_customer_date , price	2,963 , 775	Filtered order_status column to keep only "delivered" (Total removed: 2,963 rows)	Missing delivery dates make it impossible to track orders, and missing prices affect revenue analysis.
order_delivered_customer_date	8	Estimated using average delivery time	Ensure delivery time calculations are complete

- **Number of Unique Columns:** 16, at 5 main tables
- **Number of unique Orders:** 96,478

• Field	• Type
• order_id	• STRING
• order_item_id	• INTEGER
• customer_id	• STRING
• customer_unique_id	• STRING
• customer_city	• STRING
• customer_state	• STRING
• order_purchase_timestamp	• DATETIME
• order_delivered_customer_date	• DATETIME
• DeliveryDaysTime	• INTEGER
• product_id	• STRING
• ProductCategoryName	• STRING
• seller_id	• STRING
• seller_city	• STRING
• seller_state	• STRING
• price	• REAL
• review_score	• INTEGER